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Author:

Raby Hamadi

Advisors:

Dr. Hichem Besbes, Sup'Com, Tunisia

Dr. Hakim Ghazzai, Stevens Institute of Technology, NJ, USA

Dr. Yehya Massoud, Stevens Institute of Technology, NJ, USA

Work proposed and realized within:

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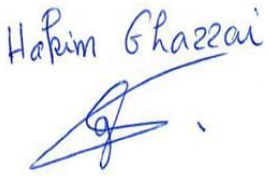
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مدينة تكنولوجيا المواصلات - طريق رواد كلم 3.5 - 2083 الغزالة أريانة - الهاتف : 000 857 71 216+ - الفاكس : 829 856 71 216+
Cité Technologique des Communications - Route de Raoued Km 3,5 - 2083 El Ghazala Ariana
Tél : +216 71 857 000 - Fax : +216 71 856 829 - E-mail : contact@supcom.tn - Web : www.supcom.tn

Research Supervisor

Dr. Hakim GHAZZAI
Research Scientist at the School of
Systems and Enterprises,
Stevens Institute of Technology, USA

Signature:



Academic Supervisor

Dr. Hichem BESBES
Professor at the Higher School of
Communication of Tunis, Supcom,
University of Carthage, TUNISIA

Signature:



Abstract

Successful portfolio management requires, in addition to advanced optimization strategies, effective recruitment of specialized financial advisors. Hiring the wrong ones can be detrimental to investors' financial goals. Without interfering with financial advisors' private portfolio optimization techniques, we design two recruitment frameworks that matches financial advisors to investors based on their profiles and features as well as the previous activities of their peers.

We first tackle the problem with an automated crowdsourcing system to organize the cooperation between financial advisors and investors. Using the database of the crowdsourcing platform, we employ an unsupervised technique to regroup advisors with high degree of similarities into clusters and hence, shrink the search space. Afterwards, we train a machine learning regression model to predict the matching score that can be achieved if an investor hires a particular advisor. These scores are converted to weights of bipartite graphs to which we apply a double phased many-to-many maximum weight matching algorithm to determine suitable investor-advisor combination. In the simulations, we investigate the performance of the proposed recruitment approach and show that, compared to other traditional approaches, higher returns can be reached for both investors and financial advisors.

As in the second part, we conceive a smart crowdsourcing platform to automatically suggest for a given investor, the potential financial advisors that are

more likely to match his/her profile and hence, provide optimized financial portfolios that responds to the investor's financial goals and preferences. To this end, an unsupervised deep learning model, namely Generative Adversarial Networks (GAN) is trained based on the previous cooperation of different pairs of investors and financial advisors using a training dataset containing pairs of an investor and the corresponding optimal financial advisor. Afterwards, the GAN is used to predict for a given investor's profile the corresponding optimal financial advisor's profile encapsulated in a gray-scale image. Then, we preform an optimized matching technique where we cluster the available advisors and define a cluster-head for each cluster. Subsequently, we assign the investor to the optimal cluster and finally assign him with the optimal advisors within that cluster. The matching process is based on the generated financial advisor profile that corresponds to the investigated investor.

Keywords: Machine learning, Deep learning, Financial technology, Crowdsourcing, Portfolio optimization, Graph theory.

Dedicated,

To you, my inspiring parents,

To my brothers,

To all those I love,

for being the pillows, role models,

cheer-leading squad and sounding boards

I have needed.

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LIST OF ACRONYMS

Artificial Intelligence (AI)

Mean Absolute Error (MAE)

Mean Squared Error (MSE)

Percentage Change Difference (PCD)

Generative Adversarial Networks (GAN)

Integer Linear Programming (ILP)

Modern Portfolio Theory (MPT)

Decision Tree Regressor (DTR)

Gradient Boosting Regressor (GBR)

Random Forest Regressor (RFR)

Portfolio Optimization Problem (POP)

General Context

0.1 Project Context

This work integrates in the obtention of the National Diploma of Telecommunication engineering from the Higher School of Communication of Tunis (Sup'Com). It was proposed and realized within Stevens Institute of Technology.

Stevens Institute of Technology is a private research university in Hoboken, New Jersey. Incorporated in 1870, it is one of the oldest technological universities in the United States and was the first college in America solely dedicated to mechanical engineering. The campus encompasses Castle Point, the highest point in Hoboken, and several other buildings around the city. Founded from an 1868 bequest from Edwin Augustus Stevens, enrollment at Stevens includes more than 5,500 undergraduate and graduate students representing 47 states and 60 countries throughout Asia, Europe and Latin America. It is classified among "R2: Doctoral Universities – High research activity". The university is home to two national Centers of Excellence as designated by the U.S. Department of Defense and U.S. Department of Homeland Security. Two members of the Stevens community, as alumni or faculty, have been awarded the Nobel Prize: Frederick Reines (class of 1939), in Physics, and Irving Langmuir (Chemistry faculty 1906–1909), in chemistry.

The main purpose of this internship was to investigate the usage of two famous concepts namely, Artificial Intelligence and Crowdsourcing to address

challenges related to financial engineering. However, and in order to keep up to date with real world problems and trending topics, we also worked on the COVID-19 topic and proposed a mechanistic model to mathematically characterize the propagation of the disease. We have also studied the challenge of computational resource allocation between internet of things devices and edge computers.

In this report, we only limit to investigate the applications of crowdsourcing and artificial intelligence and how they can be used together to solve real world challenges in financial engineering.

0.2 Background & Motivation

Modern portfolio theory (MPT) [1–4] or mean-variance analysis is one of the most challenging problems in finance. It is generally modeled as a mathematical framework to build an investment portfolio composed of multiple assets, e.g., stocks, currencies, bonds, gold, in order to maximize the expected return given a certain of levels of risk.

The MPT is developed in 1952 by the Nobel Prize winning economist Harry Markowitz [5] to study the trade-off between the expected return of a portfolio and the level of market risk. For instance, MPT can be used to construct portfolios while maximising its expected return for a given risk level and likewise, reducing the market risk level for a desired portfolio's return value. It is mainly based on the concept of diversification underpinned by concepts of risk, return, variance, and co-variance.

Diversification is a key concept to reduce the risk of a portfolio where the investment is diversified on combinations of assets that are not perfectly positively correlated. For example, if all the pairs of assets for a given portfolio have correlations of zero, it means that they are perfectly uncorrelated. However, if all the

pairs of assets for a given portfolio have correlations of one, it means that they are perfectly positively correlated. For a given portfolio, the case of all perfectly positively correlations gives the highest possible standard deviation of portfolio return. In other words, investors can reduce their exposure to individual asset risk by holding a diversified portfolio of assets. Diversification may allow to obtain, for the same portfolio expected return, a reduced risk. Therefore, with MPT the focus is no longer on the return and the risk of an asset itself but it's on the asset's contribution in the portfolio's overall risk and return [6–10]. Moreover, MPT stipulates that all the investors are risk averse, i.e., the investors are more likely to choose portfolios with lower risk for a given expected return. For instance, reaching higher expected returns will require investors to accept more risk.

Nevertheless, portfolio optimization remains a very sensitive task and requires a lot of time to survey and analyze the market in order to take the right investment decisions at the right moment. Therefore, specialized financial advisors are hired to provide efficient portfolio's management, expert recommendations, and time savings while studying the market, but for a given cost in return, e.g., a percentage of Assets under Management (AuM). Thus, hiring potential financial advisors that are worth for the amount of money that they charge is far from being a simple task especially because some financial advisors may lack the needed knowledge and skills or the expertise in some assets to reach the investors' financial planning goals even with the emergence of the robo-advisors [11, 12].

Robo advisors operates using minimum human intervention at any time and recommends investment decisions according to different mathematical algorithms/models developed by their owners. The concept of robo-advising allows investors to not only achieve their long-term financial goals but also their short term investment strategies. Based on the investors' preferences, a robo-advisor

offers automatic re-balancing strategies to manage the preferred assets in order to not only reduce the volatility of the portfolio but also improve its return. Compared to traditional advisory, robo-advisors comes with a lower cost. For example, while traditional advisors have high entry barriers, e.g., USD 250.000, automated investment with robo-advisors allows entering the market with low or even no minimum deposit. They are also accessible online or through mobile applications to track the portfolios' performances and update them at any time.

To sum up, investors need financial advisors, either humans or robo-advisors, to manage and control the assets of the portfolio in order to maximize the expected return and reduce its risk. Nevertheless, the area of expertise, the efficiency, and the MPT optimization strategies of advisors as well as the psychological characteristics of the investor may constitute important factors that highly impact the portfolio management decisions [13]. Thus, it is a must for every investor to wisely recruit the financial advisor that matches his/her strategy and preferences in order to build a portfolio that suits the targeted financial planning goals.

0.3 Related Work

In literature, several works addressed the challenge of portfolio optimization with different techniques. The authors of [14] considered a portfolio optimization model with real constraints which is commonly known as the Mean-Variance Cardinality Constrained Portfolio Optimization (MVCCPO) model and investigated two optimization algorithms to solve it, namely Multi-objective Covariance based Artificial Bee Colony and e-New Local Search based Multi-objective Optimization Algorithm. The investigated algorithms were tested on a large assets data set and the obtained results were compared with the unconstrained efficient frontier of the corresponding data set. In [15], J. Chu-xin et al.,

considered a robust portfolio model based on the Markowitz theory and turned it into a Linear matrix inequalities (LMI) with constraints. Then, the authors employed the ellipsoid algorithm and interior-point method to solve it. To validate the results the authors gave a numerical simulation in stock portfolio. Moreover, in [16] the authors studied the problem of optimal portfolio diversification where they aim to determine the structure of optimal portfolio by mathematically modeling the dynamics of the markets and the stocks. Also, in [17] the authors address the portfolio optimization challenge using the mini-max principle. Hence, they aim to minimize an objective function defined as the maximum absolute deviation of all assets which represents the portfolio's risk. M. Pieter et al., investigated the causes of the robustness of dynamic MV portfolio optimization to model misspecification errors under both the pre-commitment Mean-Variance (PCMV) and time-consistent Mean-Variance (TCMV) approaches [18]. The authors identified particular combinations of parameters that play a key role in explaining the observed model misspecification errors

Crowdsourcing is widely used in a variety of areas touching different aspects of our lives such as weather monitoring [19], event reporting [20], software engineering [21], and transportation applications [22] The most well-known crowdsourcing service investigated in finance research is crowdfunding, which consists in gathering small amounts of money from the crowd to fund a project. As example, the studies, in [23] and [24], focused on enhancing the success rate of fund raised projects by studying the reviews' sentimental effects using text mining and machine learning (ML) techniques [25, 26] or optimizing the product supply balancing the return against the risk based on the company's budget and supply perks, respectively.

On the other hand, some financial institutions resort to crowdsourcing to enhance their services. For instance, the Development Bank of Singapore and HSBC Expat have used crowdsourcing to gather ideas to improve their branch

issues and promote their brand.

Robo-Advisors are also widely investigated in literature. In [27], the authors studied the social relationships between investors and the social relationships between investors and financial institutions to propose a novel group recommender system for robo-advisor based on the topology of these financial social relationships. Also, the authors in [28] address the challenge of designing robo-advisory solutions for a specific kind of consumers, risk-averse, low-budget and inexperienced consumers. In [29], the authors focus on the actual and potential implications of AI based applications and technical issues that are related to behavioural finance. They focus on the behavioural financial decision-making process by investors and they prove that these behavioural financial decisions are important for successful execution of a customer's financial portfolios. Moreover, the typical companies in the United States (US) that manage the greatest amount of funds using robo-advisory are introduced in [30]. In this work, the author investigate US management companies applying RoboAdvisers, namely Schwab Intelligent, Wealthfront and Betterment. This work shows that these firms have the same philosophy of portfolio management through a scientific approach, but they are different in asset allocation, investment product type, and customer propensity analysis techniques.

0.4 Contributions

In this work, we address the challenge of portfolio optimization from a crowd-souring perspective while integrating two main concepts of Artificial Intelligence, namely Machine Learning and Deep learning. While other works focus on proposing new algorithms and optimization techniques of the portfolio itself, we believe that efficiently matching investors with financial advisors that

best their profiles and financial goals and preference have a direct impact on the portfolio's return. Therefore, we propose two crowd-souring-based frameworks to efficiently match investors and financial advisors.

To the best of our knowledge, our work is the first to deal with the portfolio optimization problem from a crowdsourcing framework perspective. The proposed approach does not have a direct impact on the financial advisors' algorithms optimizing portfolios. On the contrast, it considers them as private techniques that it does not aim to alter. The objective is to exploit the features and previous performances of financial advisors for a better recruitment and matching, which may consequently increase their competitiveness and hence, their quality of service.

The presented work can be divided into two major frameworks. With the two frameworks, we aim to alleviate the process of recruiting financial advisors for investors by designing intelligent crowdsourcing platforms, that aim to match suitable financial advisors, i.e., crowdsourcees or workers, to the preferences of investors, crowdsourcers or task requesters. The platforms enclose a list of financial advisors characterized by different features and hence, when a customer plans to invest, it provides its preferences, budget, as well as other parameters to the platform, which automatically recommends the suitable financial advisors.

The two proposed frameworks are the very first crowdsourcing-based platforms that integrate machine learning and deep learning algorithms to address the portfolio optimization challenge.

0.5 Report Outline

This report is divided into three chapters. In Chapter 1, we introduce the concept of crowdsourcing and speak about its main components. Afterwards, in Chapter 2, we present the very first crowdsourcing platform that matches

investors to financial advisors. In Chapter 3, GAN-based platform is presented to leverage the same task of recruiting financial advisors using more advanced technologies.

Chapter 1

Introduction to Crowdsourcing

1.1 Introduction

In this chapter, we introduce the concept of crowdsourcing. We also define the three main components that characterize it.

1.2 Crowdsourcing Concept

Crowdsourcing is a business model or concept that relies on a large group of users as third parties for outsourcing certain tasks. A crowdsourcing platform enables its users, *aka* crowdsourcers, to access to a large, relatively open and often rapidly evolving talents, skills and tangible resources, *aka* crowd, working together to achieve cumulative results through a cloud crowdsourcing platform.

This innovative sourcing model is based on the crowd's collaborative brainstorming. It is widely used to address novel challenges in different fields [31, 32]. However, in finance its applications are mainly limited to crowdfunding.

Crowdfunding is the collect of small amounts of money from a large crowd in order to fund start-ups and small businesses. Crowdfunding as a concept is not new, for instance, from a long time ago, it has been the backbone of the American political system [33–35]. Crowdfunding platforms can be dived into three categories [34]: peer-to-peer micro lending websites, funding art (books,

music, etc.) and funding new companies. The first example is peer-to-peer micro lending websites where crowdsourcers are no longer lending moneys from financial institutions but instead, from an individual or group of individuals who are willing to loan money to qualified applicants. Peerform is one of the most well-known peer-to-peer micro lending websites that was founded in 2010 by a group of Wall Street executives. With Peerform, and depending on their credit score, crowdsourcers can enjoy loans up to 25,000 USD. Second, platforms like Sellaband and TenPages are set up to support upcoming artists or writers by providing funding for the production of an album or publication of a book. Third, platforms like Crowdlender are dedicated to help entrepreneurs raise the capital they need to finance the launch or growth of their small business.

On each crowdsourcing systems four major steps are considered as shown in Fig. 1.1. In our case, first, the investor submits its request to the platform. Then, the platform recruits the most suitable financial advisors to the investor profile. Afterwards, the selected advisors may submit their proposed portfolios to the platform that share them with the investor after a certain processing, e.g., verification/validation. In this study, we focus on the recruitment process (Step 2).

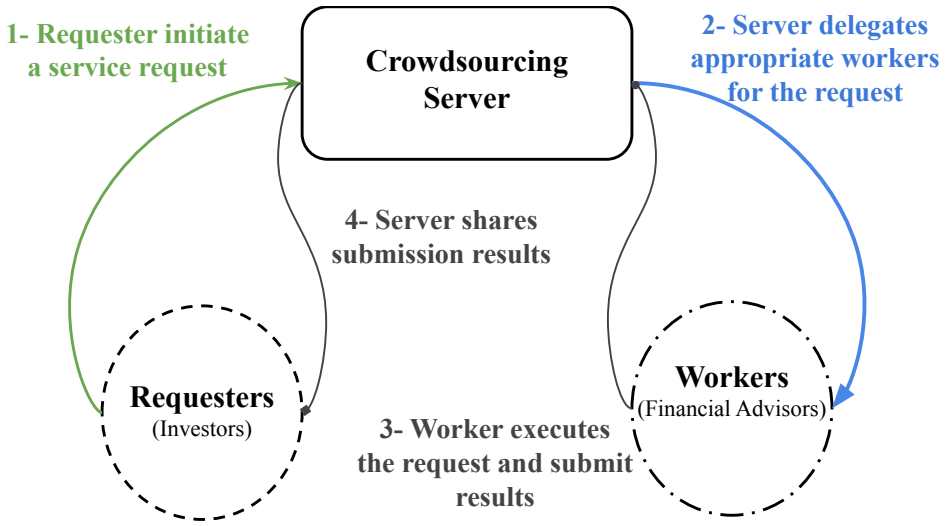


Figure 1.1: Overview of the crowdsourcing framework showing the life-cycle of an investor request.

1.3 Crowdsourcing Components

The proposed crowdsourcing system is composed of three main components: the investors, the financial advisors, and the financial crowdsourcing server.

1.3.1 Financial Advisors

In this study, we only focus on the case of human advisors. The study can be easily extended to include robo-advisors[1]. The crowdsourcing platform incorporates a data set containing the features of the financial advisors, denoted by $\mathcal{F}$. The most important features are listed as follows:

• AdvisorBN: a binary variable indicating if the financial advisor has a business number, i.e., if he is registered.

• Licence: a categorical variable indicating the financial advisors' licence name.

• LicenceBN: a binary variable indicating if the licence has a business number.

• LicenceCtrl: a numerical variable indicating the number of company(ies) or people who control the licensee's business.

• NumberDiplomas: a numerical variable indicating the financial advisors' total number of obtained diplomas.

• Experience: a numerical variable indicating the advisors' experience years.

• FailedYears: a numerical variable indicating the financial advisors' continuing professional development failure years.

• Restrictions: a categorical variable indicating the financial advisors' areas restricted from giving advice on.

• Capacity: a numerical variable indicating the total number of investors each advisors to work with simultaneously.

In addition to these features the financial advisors data set may include other

[1]The case of robo-advisors is omitted due to the non-availability of related real-world data set.

details including the areas where each financial is specialized and allowed to give recommendations on, e.g., stock markets, foreign exchange, and currencies. The data set also provides details on the qualifications and training courses the advisors have completed as well as details about their memberships of professional bodies or industry associations relevant to providing financial services.

1.3.2 Investors

The investors, denoted by $\mathcal{I}$ are one of the main components of the proposed system. The investors represent the requesters who connect to the crowdsourcing platform seeking for potential advisors to provide financial portfolios that best match their preferences and financial goals.

In this study, and since it's not possible to access investors data due to issues related to privacy and regulation authorities controlling the financial industry, the investigated investors data set is synthetically generated using the python 'faker' library to generate fake profiles.

The investigated data set represents the profile of investors that are supposed to perform activities and deal with the previously mentioned financial advisors. The data set includes the following features:

• Age: a numerical variable indicating the age of the investor.

• Gender: a categorical variable indicating the gender of the investor.

• Job: a categorical variable indicating the job of the investor.

• MaritalStatus: a binary variable indicating if the investor is married.

• Children: numerical variable indicating the number of the investor's' children.

• Risk: categorical variable indicating the risk level an investor is ready to take.

• Salary: a numerical variable indicating the salary of the investor.

• Budget: a numerical variable indicating the available budget to invest.

We statically validate the dataset to make sure it is well balanced and can

be used to take into consideration different investors profiles. For instance, as shown in the histograms of the Fig. 1.2 we can observe that the generated profiles are not equally distributed between Males (M) and Females (F). For instance the number of male investors are higher that the females as mentioned on multiple studies, for example in 2015 the percentage of female investors was less then 15% and their number was expected to rapidly increase in the future [2]. In the same figure, we can also notice that the majority of the investors have moderate risk taking levels (M) which is the case in real life. However, a fewer population prefer to take low levels of risk (L) for, obviously, a lower expected return. Furthermore, an even more fewer population are taking high risk levels (L) in order to hopefully achieve higher returns. The same observations can be remarked for the investing period where the majority of investors have moderate investing periods for a fewer population for both high and low investing periods. As for the children number, we suppose that investors have in average two children as mentioned in a statistical study about average number of own chil-

[2] `https://rb.gy/l7pcie`

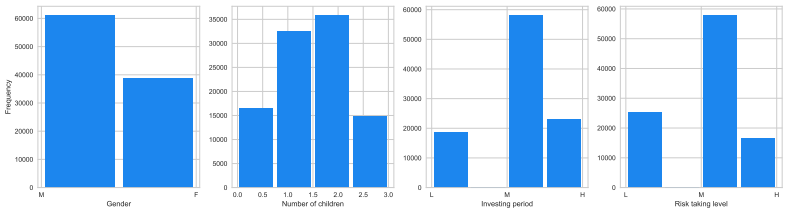


Figure 1.2: Histogram representation showing the distribution of the data set features (gender, number of children, investing period, and risk taking levels).

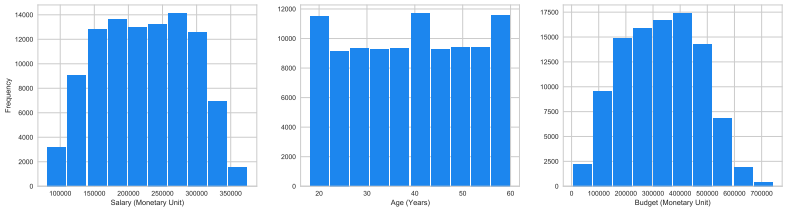


Figure 1.3: Histogram representation showing the distribution of the data set features (salary, age, and budget).

dren under eighteen in families with children in the United States[3]. In the same context, the Fig. 1.3 represents the distribution of the salary, age and budget of the investors. The investor's age is uniformly distributed to take into consideration all the age ranges. However, the salary and the budget follows a normal distribution. Hence, the Fig. 1.3 and Fig. 1.2 statistically validate the dataset distribution and shows that it contains all the possible investors' profiles. The validation of this dataset will allow us to make sure that the proposed model will be generic since it will be investigated on all the possible cases and scenarios. Also, the variety of the dataset will help us validate the performance of the framework since it will not be over-fitted on a specific scenario but on multiple and varied ones.

1.3.3 Crowdsourcing Server

The crowdsourcing server represents the backbone of a crowdsourcing system. It is a main component that links the requesters to the workers and organises the different cooperation that might occur between the two entities. The server is responsible of receiving the requests submitted by the requesters, treat them and recruit suitable workers to process the task based on a given metric. Afterwards, the server is responsible of receiving the processed task results from the selected workers, validate it and send it back to the requesters. In our work, we only focus on the recruitment part where we propose a recruitment approach to conveniently match investors with potential financial advisors in order to construct financial portfolios that best match their financial preferences and goals.

[3] https://rb.gy/ychawl

1.4 Summary

In this chapter, we presented a high level overview of the crowdsourcing concept the backbone of the the proposed solutions that will be presented in the next two chapters.

Chapter 2

Financial Advisor Recruitment: A Smart Crowdsourcing-assisted Approach

2.1 Introduction

In this chapter, we investigate the first crowdsourcing-based framework. We present the different steps of the proposed recruitment approach and the simulation results. A peudo-code of the proposed recruitment approach is given in Algorithm 1.

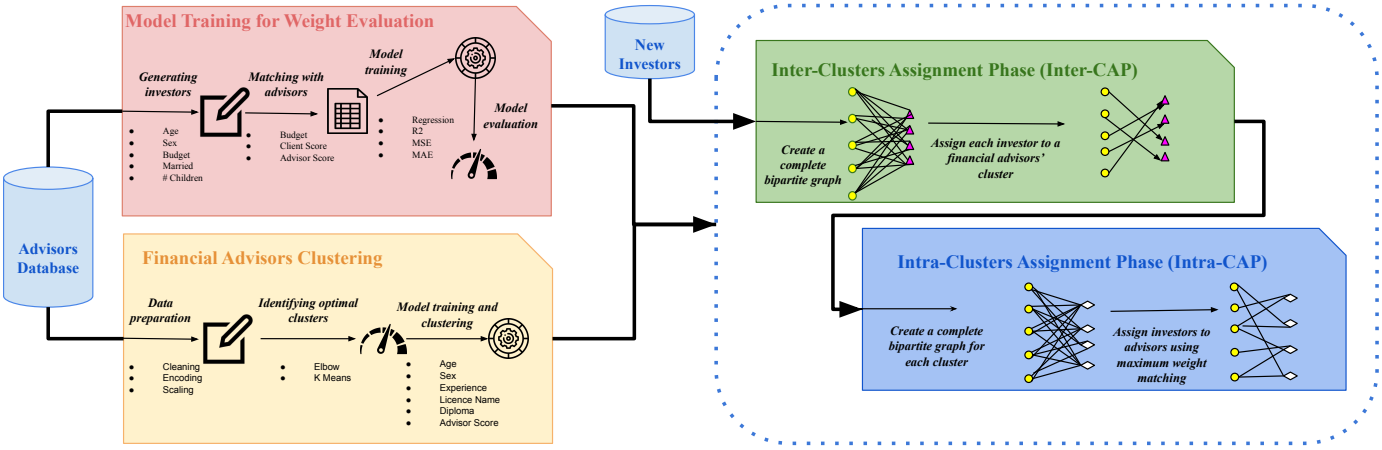


Figure 2.1: Proposed recruitment approach to match investors and potential financial advisors in a crowdsourcing platform.

2.2 System Model

We propose an intelligent recruitment framework, presented in Fig. 2.1, to assign potential financial advisors to investors, simultaneously requesting services, based on their previous experience and activities with other investors. We first cluster the available advisors in the platform using an unsupervised learning approach and represent each cluster with a cluster-head. Then, we apply a matching algorithm to a weighted bipartite graph whose nodes are formed by the cluster-heads and the investors to assign the latter to different clusters of financial advisors. Within each cluster, we individually match investors and financial advisors using a maximum weight matching with respect to two constraints, the number of financial advisors each investor can hire and the number of investors each financial advisor can support simultaneously [36]. The clustering step is added to reduce the complexity of the matching algorithm. The weights of the clusters reflect the expected return if a financial advisor A is hired by an investor B. The weights are predicted using a ML algorithm exploiting the crowdsourcing platform data set.

2.2.1 Matching Score Prediction

To measure the goodness of fit between investors and financial advisors, we assign to every pair of investor and advisor that have already cooperated a global score that reflects the return of the portfolio. The score may have positive and negative value which corresponds to a gain or a loss, respectively, but not necessarily the exact monetary gain or loss.

Given the features of a pair of investor and advisor, we predict the expected matching return of that combination based on the previous knowledge of the crowdsourcing platform. This type of prediction can be addressed using regression techniques.

In this paper, we investigate three regression models, namely Gradient Boosting Regressor (GBR), Random Forest Regressor (RFR), and Decision Tree Regressor (DTR) in predicting the score of a portfolio result of an investor-advisor combination based on the historical activities of other investors with other advisors:

• Gradient Boosting Regressor (GBR): Belonging to the family of gradient boosting algorithms, GBR showed a considerable success on different types of applications [37]. The learning algorithm starts with weak models and boosts them while training to end up with a powerful one.

• Random Forest Regressor (RFR): It belongs to the ensemble models. RFR consists of a collection of DTR which makes them stronger. The prediction is done by averaging each individual DTR prediction within the forest [38].

• Decision Tree Regressor (DTR): Decision trees are used in both classification and regression problems. They build a decision rule in the form of trees to predict a target value-based on the features affecting it [35].

2.2.2 Financial Advisors Clustering

An unsupervised clustering, particularly the K-means++ algorithm, is applied on the financial advisors set, denoted by $\mathcal{F}$, to cluster them into K groups having a high-level of similarities according to their features, e.g., assets of expertise, their qualifications, and diplomas. The K-means++ is an improved version of the K-means algorithm that ensures a smart initialization of the centroids and hence, provides an enhanced clustering results that are independent of the initialization points. On the other hand, the number of clusters is an important factor to reach a trade-off between optimizing the execution time and maintaining good matching performances. Therefore, in this paper, we address this challenge of conveniently clustering the financial advisors by using the elbow

method while considering the inertia score that represents the sum of squared distances of the investors to their closest centroids, to determine an optimized number of clusters K^*. Once the clusters are identified, a cluster-head, denoted by c_k, is determined to represent the cluster k members. Its features are obtained by averaging the normalized features of the financial advisors belonging to the cluster. The obtained cluster-head set and the corresponding sets of members are denoted by $\mathcal{C}$ and $\mathcal{F}_k, \forall k = 1, \ldots, K^*$, respectively. The main purpose of this clustering step is to shrink the large set of potential financial advisors to smaller subsets. We suppose that the clustering process is executed periodically to take into consideration the new financial advisors that may subscribe to the platform and the possible updates the enrolled advisors may apply to their profiles.

2.2.3 Investors and Advisors Matching

To assign suitable financial advisors to investors, we proceed with a double phased crowdsourcing matching approach. The first phase, called Inter-Clusters Assignment Phase (Inter-CAP), aims at assigning every investor in the investor set $\mathcal{I}$ to a cluster of workers represented by their cluster-heads $c_k \in \mathcal{C}$ with regards to the edges' weights. The second phase, called Intra-Clusters Assignment Phase (Intra-CAP), is developed to perform many-to-many assignments using bipartite graphs for each cluster k enclosing the investors and the potential financial advisors within the cluster k with regards to both investors budget and financial advisors capacities [39, 40]. The members of the assigned cluster are expected to be more suitable for the investor than their peers in other clusters.

Inter-CAP

In this phase, we only consider the cluster-head set $\mathcal{C}$ and the investors set $\mathcal{I}$. The two independent sets are inter-connected using one bipartite graph where each investor $i \in \mathcal{I}$ will be assigned to one cluster $k \in \{1, \ldots, K^*\}$ represented by a cluster-head $c_k \in \mathcal{C}$. Finally, we assign each investor i with the cluster node having the maximum weight as follows:

$$k^* = argmax_{k \in \{1,\ldots,K^*\}} \quad \Omega_{i,c_k} \forall i \in \mathcal{I}, \tag{2.1}$$

where Ω_{i,c_k} is the weight of the edge connecting the investor i and the cluster-head c_k. Its value is estimated using the ML predictor described earlier. We denote by $\mathcal{I}_k$ the set of investors associated to cluster k where $\bigcup_{k \in \{1,\ldots,K^*\}} \mathcal{I}_k = I$.

Intra-CAP

Once each investor is assigned to a cluster of financial advisors that are supposed to fit his/her profile, we end up with K^* bipartite graph representing each cluster and the investors assigned to it determined in Inter-CAP. For each graph k, the matching between the sets $\mathcal{I}_k$ and $\mathcal{F}_k$ is performed with respect to the following two constraints: the maximum number of investors each advisor is able to handle simultaneously and the number of advisor each investor is able to hire with regards to his/her available budget. In other words, the crowdsourcing platform may recommend more than one financial advisors to investors and multiple investors can be assigned to one financial advisor. In order to maximize the overall gain for all the investors, we proceed with a many-to-many matching algorithm with the objective to maximize the sum of the weights of the matched pairs expressed as $\sum_{i \in \mathcal{I}_k} \sum_{f \in \mathcal{F}_k} \epsilon_{i,f} \omega_{i,f}, \forall k \in \{1, \ldots, K^*\}$ where $\omega_{i,f}$ is the weight connecting an investor i to a financial advisor f. Its value is predicted using the ML approach. The variable $\epsilon_{i,f}$ is a binary variable to be determined

by the matching algorithm. It is equal to one if investor i is assigned to advisor f and zero, otherwise.

The recruitment approach is finalized by recommending a list of financial advisors to each investor. In practice, the latter will have the freedom to choose how many and which financial advisors to hire from that list.

Algorithm 1 Proposed Algorithm for Financial Advisors Recruitment

0: **Input:** Set of investors $\mathcal{I}$, set of advisors $\mathcal{F}$.
0: Find K^* the optimal number of clusters using elbow method.
0: Apply K-means algorithm with $K = K^*$ on $\mathcal{F}$ and determine the cluster-head set $\mathcal{C}$.
0: Build a bipartite graph to inter-connect the elements of $\mathcal{I}$ and $\mathcal{C}$. The corresponding weights $\Omega_{i,k}$ are predicted using the ML model.
0: Match each investor to a cluster $k \in \{1, \ldots, K^*\}$ using (2.1).
0: **for** $k \in \{1, \ldots, K^*\}$ **do**
0: Create a complete bipartite graph for each cluster k with the investors $\mathcal{I}_k$ and workers $\mathcal{F}_k$ assigned to that cluster. The corresponding weights $\omega_{i,f}$ are predicted using the ML model.
0: Apply the many-to-many matching algorithm while maximizing the sum of weights of the selected pairs.
0: **end for**=0

2.3 Results & Discussions

In this section, we apply the proposed recruitment approach on a real-world data set where 300 financial advisors' profiles exist. The latter is obtained from the Australian Securities and Investment Commission financial advisers register.

Notice that the data set does not include any details about the algorithms or strategies adopted by the advisors to optimize the portfolios. The elbow method used with the K-means clustering algorithms identifies $K^* = 9$ clusters of financial advisors.

On the other hand, the investors data set composed of 500 profiles is created using synthetic data to generate diverse and random investors' profiles.

We suppose that the psychological risk taking levels of each investor affects

the portfolio assets within the portfolio and as a result its expected return.

Afterwards, a training history of different investor-advisor combinations is generated to train the ML models. Once trained, the models predict the expected score of a portfolio for an investor when taking instructions from a specific advisor

We employ the Mean Squared Error (MSE), Mean Absolute Error (MAE), Percentage Change Difference (PCD), and the coefficient of determination R^2 as evaluation metrics of our models' performances. They are expressed as follows:

$$MSE = \sum_{i=1}^{N} \frac{(y_n - \hat{y}_n)^2}{N}, \quad MAE = \sum_{i=1}^{N} \frac{|y_n - \hat{y}_n|}{N}, \tag{2.2}$$

$$PCD = \sum_{i=1}^{N} \frac{|y_n - \hat{y}_n|}{\left(\frac{y_n + \hat{y}_n}{2}\right)}, \quad R^2 = 1 - \frac{\sum_{i=1}^{N} (y_n - \hat{y}_n)^2}{\sum_{i=1}^{N} (y_n - \overline{y})^2}, \tag{2.3}$$

where N is the total number of the testing cases, y_n, $\overline{y}$, and $\hat{y}_n$ are the real measures, the average of all real measures, and the predicted overall return of the portfolio, respectively. While MAE, MSE, and PCD are giving insights about the loss on testing data, R^2 is used to evaluate the goodness of fit of the algorithms, i.e., it tells how well are the predicted values in approximating the real ones,

The performances of the ML models previously discussed are summarized in Table 2.1. The GBR model is shown to be outperforming the other machine learning candidates which is predictable since the GBR is built on top of the other two models. It is also able to effectively predict the matching scores with an MSE and R^2 close to 0 and 1, respectively. In this study, we investigate the performance of these three algorithms and select the best that fits our data set and apply it to the next Phase. It is worth to note that the selected algorithm is

Table 2.1: Performances of the different machine learners for edge weight predictions

Model	**Train. Loss**	**Test. Loss**	**MSE**	**MAE**	**PCD**	R^2
DTR	2.508	2.510	0.106	0.606	0.965	0.910
RFR	0.444	0.516	0.055	0.161	0.516	0.981
GBR	0.086	0.333	0.002	0.039	0.333	0.992

not necessarily the most suitable one for all data sets. Indeed, we can not claim the global superiority of an algorithm over the others. In the sequel, we employ the GBR algorithm.

We compare our proposed matching method with two other practical matching techniques. A random matching where an investor is randomly matched to an advisor, which is potentially close to traditional scenarios, and a first in first out (FIFO) matching where the first investor connected to the platform is associated to the available advisors that provide the highest matching scores. For fair comparison, the experiments are performed on a 1000 financial advisors where 1700 tests are realized. At each realization, an investor is matched to a financial advisor according to the matching algorithm. In Fig. 2.2, we plot the histograms of the expected return of the same 1700 realizations for each matching technique and, in Table 2.2, we provide the corresponding statistical details. It is clear that our proposed matching algorithm outperforms the other approaches in providing matching results that maximize the average return of all the investors with a minor deviation as reflected by the obtained standard deviation (1.12 for the proposed approach versus 2.26 and 3.92 for the FIFO and random matching approaches, respectively). These results confirm that we are able to reduce the gap between the minimum and maximum matching scores. Hence, with a dedicated recruitment strategy, higher monetary gains can be achieved for both investors and financial advisors.

In Table 2.3, we illustrate the outputs of the different steps of the proposed recruitment algorithm including the number of advisors and associated investors

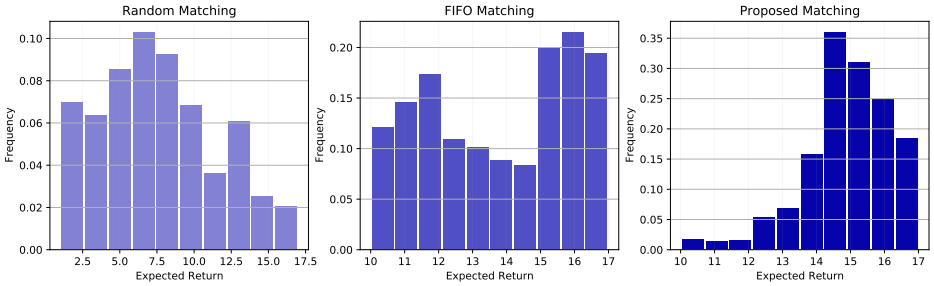


Figure 2.2: Statistical comparison of the proposed matching algorithm vs. two basic matching techniques.

Table 2.2: Comparison of the proposed matching performances' with random and FIFO matching

Matching Tech.	**Av. Score (%)**	**Max. Score (%)**	**Min. Score (%)**	**Stand. Dev.**
Proposed Matching	14.88	16.99	10.01	1.12
FIFO Matching	12.34	16.85	3.11	2.26
Random Matching	7.59	16.99	1.09	3.92

per cluster as well as some statistics about the achieved matching scores. For example, 45 investors are assigned to cluster zero which contains 38 advisors, while cluster 3 is formed by 7 advisors, which may serve 68 investors. We notice also that the average score for investors for all clusters are relatively close to each other, which reflects the efficiency of the matching algorithm in associating investors to their relevant advisors. Moreover, the statistics show a small gap between the minimum and the maximum matching scores per cluster.

Finally, Table 2.4 presents examples of individual assignments. Investor with ID 112 has been assigned to three potential financial advisors with expected scores varying between 11.5% and 19.86%, while investor 7 is associated to a single advisor providing a score of $\approx 17\%$. The framework adapts the financial advisor recommendation according to the criteria imposed by the investor and their budgets.

Table 2.3: Statistics of the advisor-investor assignment using the proposed recruitment approach

Cluster ID	#Adv.	#Inv.	Min. Score (%)	Max. Score (%)	Av. Score (%)
0	38	45	12.586	18.230	14.425
1	114	64	22.985	28.903	25.365
2	8	58	12.234	17.708	15.389
3	7	68	18.658	23.023	20.457
4	26	47	13.001	16.648	14.589
5	13	64	14.298	19.493	17.256
6	35	47	15.013	22.753	18.235
7	15	55	10.170	12.168	11.698
8	44	52	16.698	23.058	19.654

Table 2.4: Examples of individual matching cases

Inv. ID	Assigned Adv.	1st Adv. Score (%)	2nd Adv. Score (%)	3rd Adv. Score (%)
112	3	19.860	12.275	11.568
96	2	15.245	15.213	-
7	1	17.103	-	-

2.4 Summary

In this chapter, we have presented the proposed double-phased recruitment approach to efficiently manage portfolio optimization problems. We mainly focused on the recruitment phase to determine the financial advisors that may offer the best portfolio optimization solutions.

Chapter 3

Financial Advisor Recruitment: A Smart GAN based Approach

3.1 Introduction

In this chapter, we investigate the second crowdsourcing-based framework. We present the different steps of the proposed recruitment approach and the simulation results.

3.2 System Model

In this section, we represent the work-flow of the proposed framework illustrated in Fig. 3.1. First, investors initiate the system by submitting requests to

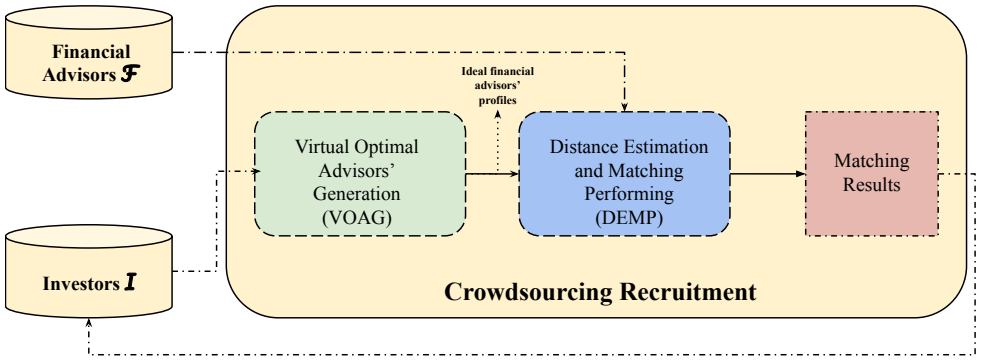


Figure 3.1: A high-level overview of the proposed smart crowdsourcing recruitment approach.

the server. Then, two phases are operated. The first phase, called Virtual Optimal Advisors' Generation (VOAG) phase, employs the GAN to generate, for each investor, a virtual financial advisor's profile that is expected to best fit the requirement of the investor given its profile data and demand. The output of the VOAG phase is employed to determine existing financial advisors that provide closest features to ideal virtual financial advisor generated by the GAN. Afterwards, we apply the second phase, called Distance Estimation and Matching performing (DEMP) phase, to assign available and appropriate financial advisors to investors.

3.2.1 Financial Advisors' Signature Encoding

Based on the previously mentioned features provided for each financial advisor, we characterize each financial advisor's profile with a signature in the form of a gray-scale image representing all its features. This characterization of financial advisors' profiles will be employed to distinguish between existing profiles and will be used as an input to GAN in the next phase to generate ideal financial advisors' profiles associated to each POP requester. By integrating the AI-based image processing technique, we propose to simplify the recruitment process and enable rapid recognition of suitable financial advisors existing in the platform to different investors' profiles.

The shape of the gray-scale images depends on the number of features provided on the data set. For instance, to define a unified shape, if a data set provides features that can not be reshaped to the desired dimensions, features with low significance and low entropy can be discarded. Accordingly, if the number of features is below the desired dimensions the gray-scale image can be extended with zero pixels. In Fig. 3.2, we provide two examples of financial advisors signatures. The database include many human financial advisors' pro-

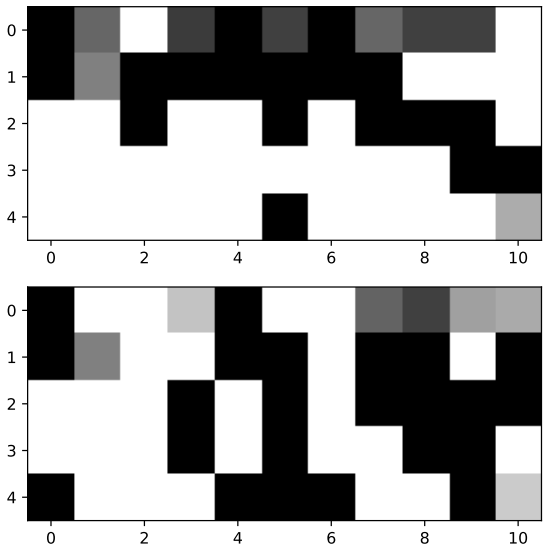


Figure 3.2: Outputs of the financial advisors' profiles encoding: financial advisor's signatures.

files characterized by 55 features. After normalization, we convert the features vectors into two dimensional matrix which is itself converted to a gray-scale image. Hence, each financial advisor will be characterized by its gray-scale image or signature that will be used to train the GAN model employed in the VOAG phase.

3.2.2 Virtual Optimal Advisors' Generation (VOAG) Phase

In this study, we suppose that the recruitment process is performed with regard to the features of the investor (i.e., crowdsourcing requester) and the financial advisor (i.e., crowdsourcing worker). The objective is to empower the crowdsourcing platform with smart and automated tool that automatically finds the "ideal" financial advisor profile that matches the need and the profile of the requester. Therefore, we aim to generate, based on the features of the investor given as an input, the ideal financial profile that best match the investor's profile encapsulated in a gray-scale image that represents its signature.

We propose a solution based on the generative modeling approach [41–45]

named Generative Adversarial Networks (GAN) [46–48]. In general, generative modeling are unsupervised AI algorithms that automatically discover the features, patterns, and regularities of an input data set in order to be able to independently output, based on random noises, samples that plausibly could have extracted from the original input data set. However, GAN is an improved generative model such that training the network is formulated as a supervised learning problem where two sub-models are involved, namely a Generator ($\mathcal{G}$) and a discriminator ($\mathcal{D}$). The generator is trained to transform a random noise input to a desired output form. In contrast, the discriminator tries to classify the generated samples into two classes, fake images (comes from the generator) and real (comes from domain data set). The training process is an adversarial game between $\mathcal{G}$ and $\mathcal{D}$ where both models are trained simultaneously. The game ends when the $\mathcal{G}$ model is able to 'fool' the $\mathcal{D}$ model, i.e., $\mathcal{G}$ is generating plausible examples.

GAN is an exciting and rapidly changing field, therefore it is widely used to address multiple challenges and applications, particularly in computer vision [49–52]. GAN is also widely used in financial engineering. For example, the authors of [53] studied the stock market prediction problem using GAN. The model was built using a Multi-Layer Perceptron (MLP) as a generator and the Long Short-Term Memory (LSTM) as the discriminator. The network is trained to forecast the daily closing price of several stocks based on the S&P 500 Index. The authors showed that promising results can be obtained using GAN compared to other deep learning and machine learning algorithms.

To train the GAN model, we proceed by a simultaneous training cycle for both $\mathcal{G}$ and $\mathcal{D}$. The discriminator is trained to distinguish between real images and fake images generated by $\mathcal{G}$. On the other hand, we aim to train the $\mathcal{G}$ based on the feedback of $\mathcal{D}$, i.e., the generator weights will be updated according to the performance of the discriminator. Recall that the objective is to

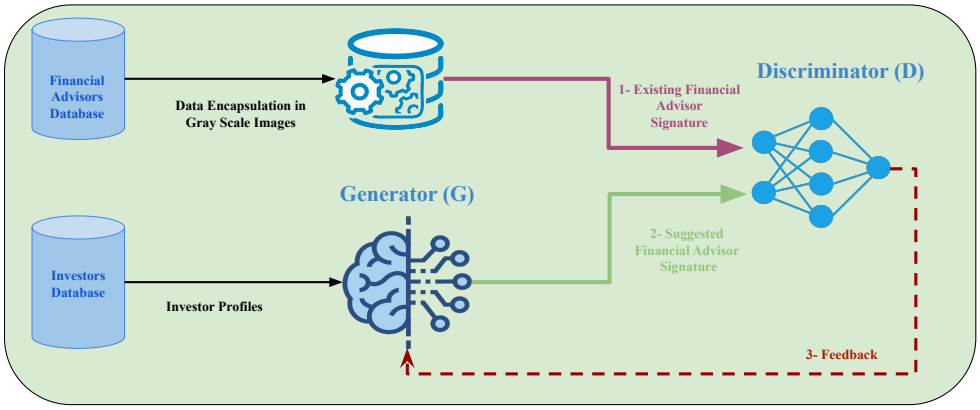


Figure 3.3: Standalone model stacking and training cycle of the proposed GAN for generating ideal financial advisors profiles.

generate ”ideal” financial advisors’ profiles encapsulated in gray-scale images based on the profile of the investor of interest. We denote by ’real image’ an existing financial advisors’ signature and with ’fake image’ a signature generated by the GAN given an investor profile. For instance, if the discriminator correctly classifies the fake image, the generators’ weights will be updated and it will learn the features and characteristics of the real financial advisors’ signatures. On the contrary, if the discriminator is wrongly predicting fake images the feedback will be poor and hence, the generators’ weights will not be updated conveniently as no new information are learnt on how to construct close to reality images. Therefore, an adversarial relationship is established between $\mathcal{D}$ and $\mathcal{G}$. Therefore, the GAN model is defined as the sequential stacking of both the generator and the discriminator such that we input the investor profile to $\mathcal{G}$. The latter will generate fake samples that are directly fed to $\mathcal{D}$ which will classify them and output a feedback that is going to be used to regulate the generator’s weights. Accordingly, the generator will be trained via the training of the logical GAN model.

The GAN model is trained while minimizing an objective function derived from the cross-entropy between the real and generated distributions defined as

follows:

$$L = \min_{x} \max_{z} E_x[\log(\mathcal{D}(x))] + E_z[\log(1 - \mathcal{D}(\mathcal{G}(z)))], \tag{3.1}$$

where

• $\mathcal{D}(x)$ is the discriminator's estimate of the probability that a real financial advisor signature x is real.

• E_x is the expected value over all real data instances x.

• $\mathcal{G}(z)$ is the generator's output given an investor profile z.

• $\mathcal{D}(\mathcal{G}(z))$ is the discriminator's estimate of the probability that a generated financial advisor signature is real.

• E_z is the expected value over all generated fake instances $\mathcal{G}(z)$.

In this study, to define the GAN model, we first define the discriminator $\mathcal{D}$ and the generator $\mathcal{G}$ as two standalone models. Each of these two models is an independent model and hence, the GAN is a sequential model results of the stacking of the two models.

The Discriminator $\mathcal{D}$

The discriminator $\mathcal{D}$ model is defined as a sequential deep learning model where the first two layers are a two-dimensional convolution layer separated by a dropout layer to prevent over-fitting. The first two layers are followed by a leaky version of a Rectified Linear Unit layer (LeakyRelu) and another dropout layer. Finally, the output layer is a dense layer with one node activated by the 'sigmoid' activation function to predict whether the input sample is real or fake. The sigmoid function, denoted by ζ, is defined as follows:

$$\zeta(x) = \frac{1}{1 + \exp^{-x}} \tag{3.2}$$

The discriminator $\mathcal{D}$ is trained to minimize the binary cross entropy loss function, appropriate for binary classification. The binary cross entropy loss function is the preferred loss function under the inference framework of maximum likelihood and it is defined as follows:

$$Loss_D = (y - 1) \times \log(1 - \hat{y}) - y \times \log \hat{y} \tag{3.3}$$

The Generator $\mathcal{G}$

We define the generator $\mathcal{G}$ as a sequential deep learning model where the first layer is a dense layer having as an input dimension the number of features that characterizes the investors. The dense layer is the followed by a two-dimensional convolution layer and a reshape layer to reshape the data into the desired output dimensions. Finally, another two two-dimensional convolution layers are added to the $\mathcal{G}$ model to output the desired financial advisors' signature.

On the training process, the $\mathcal{G}$ model tries to minimize L by minimizing the term $\log(1-\mathcal{D}(\mathcal{G}(z)))$, i.e., by generating, given a random noise z, a fake sample $\mathcal{G}(z)$ that shows the same patterns of the real data such that $\mathcal{D}$ classifies it as real, and hence $\mathcal{D}(\mathcal{G}(z)) = 1$. In contrast, $\mathcal{D}$ tries to maximize L by classifying all the real samples x and the fake samples $\mathcal{G}(z)$ as real and fake, respectively, i.e., $\mathcal{D}(x) \to 1$ and $\mathcal{D}(\mathcal{G}(z)) \to 0$.

After convergence, the generator will be able to generate ideal financial advisors' profile encapsulated in a gray-scale image based on the features of an investor profile given as an input

3.2.3 Distance Estimation and Matching Performing (DEMP) Phase

The proposed framework aims not only to find the ideal financial advisor that matches an investor profile but also to determine closest profiles existing in the database to this ideal virtual profile. Moreover, our framework aims to maximize the overall gain for a group of investors seeking services from a group of financial advisors. Indeed, some financial advisors might match the profiles of multiple investors however, they cannot serve them simultaneously. On the other hand, some investors may require a portfolio management service from multiple financial advisors to diversify their investment and maximize their gain. Therefore, in this phase, we propose to design a dedicated many-to-many investor-to-advisor assignment to enhance the overall matching output while overcoming these challenges.

We suppose that each investor $i \in \mathcal{I}$ is only able to recruit n_i financial give his/her budget. On the other hand, each financial advisor $f \in \mathcal{F}$ is only able to work with m_i investors simultaneously.

The matching problem can be then formulated using an Integer Linear Programming (ILP) as follows:

$$\max_{x_{i,j}\in\{0,1\}} \sum_i \sum_j w_{ij} \times x_{ij}$$
$$\text{s.t.} \quad \begin{cases} \sum_i x_{ij} \leq n_j, & \forall j \in \{1,\ldots,\mathcal{F}\}, \\ \sum_j x_{ij} \leq m_i, & \forall i \in \{1,\ldots,\mathcal{I}\}, \\ \sum_j x_{ij} \geq 1, & \forall i \in \{1,\ldots,\mathcal{I}\}, \end{cases} \tag{3.4}$$

where x_{ij} is a binary decision variable indicating whether a financial advisor j is assigned to investor i. The parameter w_{ij} denotes the similarity between the ideal signature generated by the GAN for the profile of investor i, ob-

tained from the VOAG phase, and the signature of financial advisor j existing in the database. The value of $w_{i,j}$ is defined as the Structural Similarity Index (SSIM) [54, 55] and can be computed as follows:

$$w_{ij} = \frac{(2\mu_i\mu_j + \alpha_1) \times (2\sigma_{ij} + \alpha_2)}{(\mu_i^2 + \mu_j^2 + +\alpha_1) \times (\sigma_i^2 + \sigma_j^2 + \alpha_2)}, \tag{3.5}$$

where μ_i, μ_j, σ_i^2, σ_j^2, and σ_{ij} are the average, variance, and co-variance of signatures associated to i and j, respectively. Moreover, α_1 and α_2 are two stabilization variables to stabilize the division with weak denominator and are defined as $(k_1Q)^2$ and $(k_2Q)^2$, respectively. By default, $k_1 = 0.01$, $k_2 = 0.03$ and $Q = 2^p - 1$, where p is the number of bits on each pixel. In our case, $p = 8$ bits since we are employing gray-scale images and each pixels' value is ranged between 0 and 255 hence, 2^8=256 possible values exist.

The first constraint in (3.4) indicates that a financial advisor j cannot serve more than n_j investors. The second constraint indicates that an investor i is willing to recruit a maximum of m_i financial advisors. Finally, the third constraint is added to ensure that all the investors are assigned to at least one financial advisor.

The problem formulated in (3.4) is an NP-hard ILP that can be solved optimally using very complex methods such as the exhaustive search or the branch-and-bound method. Off-the self software incorporating these algorithms can be employed to this end. However, in practice, and for real-time services, it is highly recommended to employ low complexity algorithms, For this reason, we propose to model the problem as a weighted bipartite graph composed of the two sets $\mathcal{I}$ and $\mathcal{F}$. The edge interconnecting the vertices of the two sets are weighted with the similarity metric $w_{i,j}$. Afterwards, we employ a constrained maximum weight matching algorithm to perform the matching process by selecting the edges that maximize the overall similarity between the suggested financial ad-

visor profile for each investor and the existing financial advisors' profiles with respect to the capacity constraints.

For a large-scale database, and as an optional step, we propose to split the matching process into two steps. In the first step we cluster the available financial advisors using an unsupervised clustering algorithm into K clusters and select a cluster-head for each cluster. The cluster-head is selected to represent all members of its cluster and its features are selected by averaging over all the features of the other members of the same cluster. Note that for clustering multiple algorithms can be used K-Means, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), Mean-Shift, etc. Afterwards, we assign the investors to the best financial advisors' cluster by approximating the similarity of the suggested financial advisors profiles with the cluster-heads. This step only considers the investors $\mathcal{I}$ and the cluster-head set denoted by $\mathcal{C}$. Once each investor is assigned to a financial advisors cluster, the second phase aims to assign the investors within the clusters. To this end, the two sets are interconnected with a bipartite graph where the weights are computed as the similarity of the generated signature that best matches each investor profile $\mathcal{G}(i), \quad \forall i \in \mathcal{I}$ and the signatures of all the cluster-heads. Afterwards, we perform maximum weight matching to assign each investor $i \in \mathcal{I}$ to a cluster $k \in \{1, \ldots, K\}$ represented by a cluster-head $c_k \in \mathcal{C}$ as follows:

$$k = argmax_{k \in \{1,\ldots,K\}} \quad \omega i, c_k \quad \forall i \in \mathcal{I}, \tag{3.6}$$

where $\omega i, c_k$ is the weight of the edge connecting the investor i and the cluster-head c_k. We denote by $\mathcal{I}_k$ the set of investors associated to cluster k where $\bigcup_{k \in \{1,\ldots,K\}} \mathcal{I}_k = \mathcal{I}$.

Afterwards, we operate the intra-clusters matching where we consider the investors I_k assigned to the cluster of advisors $k \in \{1, \ldots, K\}$. For each cluster,

we perform the same maximum weight matching to finally assign the investors to the corresponding optimal financial advisors based on the SSIM.

Recall that the clustering step is optional and we can directly use the full set of $\mathcal{F}$ in the bipartite matching process. In the next section, the impact of clustering will be investigated.

3.3 Results and Discussion

3.3.1 Generative Adversarial Networks' Training Process

In this paper, we extrapolate the challenge of recruiting financial advisors to an image generation task based on random noise. The main idea of our work is employ trending artificial intelligence techniques, mainly deep neural networks and particularly, CNNs to generate, based on a random noise that represents an investors' profile, the optimal financial advisors profile encapsulated on a gray-scale image that best matches the characteristics of the investor given as an input. To this end, we train a GAN to automatically predict, based on the profile of each investor, the corresponding optimal financial advisors' profile. Thus, we construct a database containing pairs of financial advisors and investors. The training data set contains $N = 412448$ rows, each row of this data set contains the investor profile and the associated 'ideal' financial advisors' signature that corresponds to that investor. Hence, the training data set of the GAN is composed of N pairs of $(i, f_i^*) \in \mathcal{I} \times \mathcal{F}$, where i the investor profile and f_i^* is the ideal financial advisors' profile associated to investor i. The ideal advisor for each investor is determined using an algorithm proposed in [56].

The training of the GANs is performed as follows: first, the investor profile is fed as a noise to the generator $\mathcal{G}$. Then, based on this noise G generates a two dimensional gray-scale image that represents the financial advisors signa-

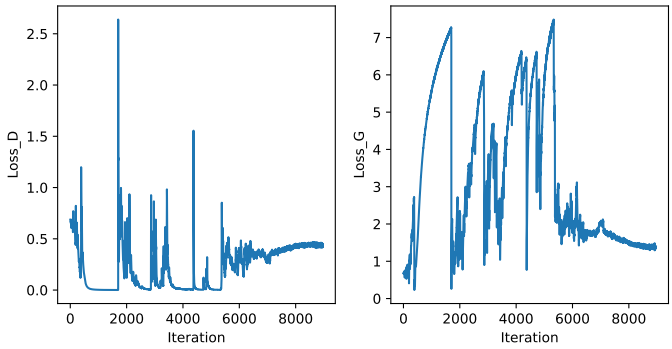


Figure 3.4: Discriminator $\mathcal{D}$ and Generator $\mathcal{G}$ loss evolution for each iteration of the training process

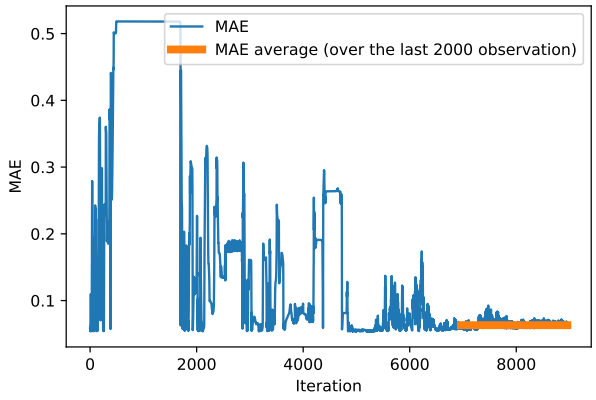


Figure 3.5: Mean Absolute Error variation of the performed tests on each iteration of the training process

ture. Afterwards, the generated image is fed to the discriminator $\mathcal{D}$. The latter classifies the image as real if the image is close to the targeted financial advisor's signature or False if not. Then, the classification result will be forwarded as a feedback to $\mathcal{G}$. In his turn $\mathcal{G}$ will use the provided feedback to update his weights, and hence after thousands of iterations, G learns how to automatically outputs the optimal financial advisors' profile based on a random investors' profile fed as a noise input. The GAN training process was done over $e = 70$ epochs and the data set was divided on $b = 3200$ batches and hence, for each epoch $i_e = \frac{N}{b} = 128$ iterations are performed for a total $i = e \times i_e = 8960$ iterations on the training process. Fig. 3.3 summarizes the training cycle of the GAN where for each iteration, we start by selecting random real samples from the input data set and generating fake samples using the investors profiles. Then, both fake and real samples are stacked and fed to the $\mathcal{D}$ model to train it and hence, update its weights. Afterwards, we also train the $\mathcal{G}$ model via the discriminator's error

The evolution of the loss function of the two models is represented in Fig. 3.4. In this figure, the adversarial relationship of the two models is shown. For instance, when the $\mathcal{D}$ is fooled by the fake images generated by $\mathcal{G}$, i.e., the loss function of $\mathcal{D}$ is high, we can see that the loss function of $\mathcal{G}$ is low and vice-versa. after 8000 iterations the two models reach stability and both loss functions converge to a stable value. To better prove the convergence of the GAN, we perform at the end of each iteration tests on a sample of 500 combinations of investors and financial advisors. For instance, after the end of each iteration we use the partially trained $\mathcal{G}$ to generate optimal advisors' signatures for the test samples. Then, we calculate the pixel-wise Mean Absolute Error (MAE) between the generated and the real advisors signatures. The obtained results are represented in Fig. 3.5. Figure shows that the generator's performance is evolving with the iterations as the MAE is decreasing. At the end of the training

period, the MAE reaches very low values indicating that the $\mathcal{G}$ is become able to generate optimal financial advisors profiles encapsulated in gray-scale images to define their signatures. Hence, based on the profile of each investor given as an input the generator is able to accurately generate the optimal financial advisors' signature that best match the investors profile.

Many-to-Many Bipartite Matching

As previously mentioned the matching is divided into two phases. The first phase only considers the investors $\mathcal{I}$ and the cluster-heads $\mathcal{C}$. The two sets are interconnected with a bipartite graph where the weights are computed as the similarity of the generated signature that best matches each investor profile $\mathcal{G}(i) \quad \forall i \in \mathcal{I}$ and the signature of the cluster-heads $c_k \in \mathcal{C}$. Afterwards, we perform maximum weight matching to assign each investor $i \in \mathcal{I}$ to a cluster $k \in \{1, \ldots, K^*\}$ represented by a cluster-head $c_k \in \mathcal{C}$ as follows:

$$k^* = \max_{k \in \{1,\ldots,K^*\}} w_{i,c_k} \forall i \in \mathcal{I}, \tag{3.7}$$

where wi, c_k is the weight of the edge connecting the investor i and the cluster-head c_k. We denote by $\mathcal{I}_k$ the set of investors associated to cluster k where $\bigcup_{k \in \{1,\ldots,K^*\}} \mathcal{I}_k = I$.

Afterwards, we operate the intra-clusters matching where we consider the investors I_k assigned to the cluster of advisors $k \in \{1, \ldots, K^*\}$. For each cluster, we perform the same maximum weight matching to finally assign the investors to the corresponding optimal financial advisors based on the $SSIM$.

3.3.2 Matching Results Validation

To validate the proposed matching results' performances, we compare the obtained results with other results obtained with the ILP resolved using Guropi

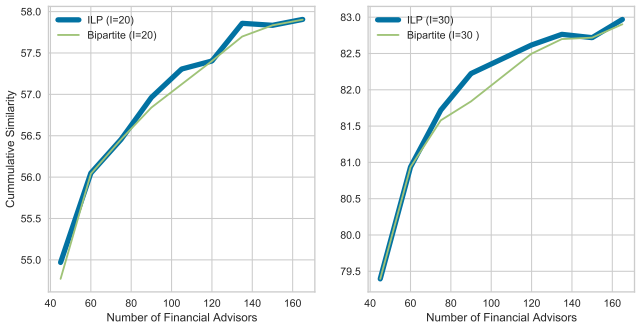


Figure 3.6: Performance comparison of the proposed matching technique to get the optimal results.

We perform two tests on 20 and 30 investors, respectively. For each test, we operate the matching algorithm to match the investors while varying the number of financial advisors to define multiple scenarios, i.e., each scenario is defined with the initially selected investors and the randomly selected financial advisors. For fair comparison, each scenario is repeated with different advisors multiple times and we plot the averaged result in Fig. 3.6.

3.4 Summary

In this final chapter, we presented our simulations and results for different use cases. The behavior of the autonomous UAV for the investigated scenarios corroborates the ability of our frameworks to establish efficient employment of UAVs in smart cities applications.

Conclusion and Future Work

Portfolio optimization is a very important and sensitive task and in literature lot of studies are addressing this challenge. In this project, we propose the very first solution that aims to maximize financial portfolio's returns by focusing on the recruitment of financial advisors. The proposed solutions study the portfolio optimization task from a crowdsourcing perspective where the focus is mainly on the recruitment process.

In the first study, we proposed a crowdsourcing system to efficiently manage portfolio optimization problems by focusing on the recruitment phase to determine the financial advisors that may offer the best portfolio optimization solutions. We have employed machine learning to predict the expected score of a matching resulting of a combination of an investor and a financial advisor based on their profiles and previous activities.

In the the second study, we extrapolated the recruitment task to an image processing task where we employ Generative Adversarial Networks to address it. The proposed solution is investigated on the case of human financial advisors but can be easily extended to include robo-advisors once related data sets are available.

Without interacting with the financial advisors working strategies, preliminary results have shown, that the framework can help increase the competitiveness among financial advisors and hence improve the quality of service and benefits for the customers. The simulations results showed that our solution can

provide accurate matching of investors to potential financial advisors resulting in portfolio's with higher returns and hence, an important gain to both entities.

In our future work, we plan to enhance our platforms with other frameworks to better automate the recruitment, submission verification, and privacy management processes. We also aim to generalize the platforms to include not only traditional financial advisors but also robo-advisors and compare the performances of these two entities in terms of the investors' gain and profits. We also aim to include safety measures to protect the privacy of both investors and financial advisors subscribed to our platforms.

Accepted/Submitted publications

[1] A. Khanfor, **R. Hamadi**, H. Ghazzai, Y. Yang, M. R. Haider and Y. Massoud, *"Computational Resource Allocation for Edge Computing in Social Internet-of-Things"* **accepted** *in IEEE 63nd International Midwest Symposium on Circuits and Systems (MWSCAS'20)*, Springfield, MA, USA.

[2] **R. Hamadi**, H. Ghazzai, H. Besbes, and Y. Massoud, *"Financial Advisor Recruitment: A Smart Crowdsourcing-assisted Approach"* **accepted** *in IEEE Transactions on Computational Social Systems.*

[3] H. Friji, **R. Hamadi**, H. Ghazzai, H. Besbes, and Y. Massoud, *"A Generalized Mechanistic Model for Assessing and Forecasting the Spread of the COVID-19 Pandemic"* **submitted** *to IEEE Access*.

[4] **R. Hamadi**, H. Ghazzai, H. Besbes and Y. Massoud, *"Financial Advisor Recruitment: A smart GAN based Approach"* **to be submitted**

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